

IST604 – Web Services & Distributed Computing

Chapter 3: Developing Web Services Using SOAP — Practice MCQ

Instructions: Select the best answer. Answers are provided at the end.

Section A: SOAP Transmission & Standards

1. In SOAP message transmission, what is the role of a SOAP Intermediary?

- A) The final destination that processes the SOAP message
 - B) The node that creates and initiates the SOAP message
 - C) An optional node between sender and receiver that can perform filtering, logging, or caching
 - D) A registry that stores service descriptions
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2. What is the term for the final intended destination of a SOAP message?

- A) SOAP Broker
 - B) SOAP Relay
 - C) SOAP Router
 - D) Ultimate SOAP Receiver
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3. Which of the following best describes the role of WSDL in a SOAP-based web services ecosystem?

- A) It transports SOAP messages between service endpoints
 - B) It acts as the metadata language describing what the service offers, where it is, and how to access it
 - C) It registers service providers in a global directory
 - D) It encrypts SOAP messages for secure transmission
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4. UDDI registries store web service interface descriptions using which language/format?

- A) JSON
 - B) OpenAPI/Swagger
 - C) WSDL
 - D) REST endpoints
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Section B: Implementation Structure

5. In JAX-WS best practice, what does SEI stand for and what is its role?

- A) Service Execution Instance — runs the web service operations
 - B) Service Endpoint Interface — declares the methods that are the web service operations
 - C) SOAP Encoding Interface — defines the XML encoding rules
 - D) Service Exchange Interface — manages client-server communication
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6. What does SIB stand for in the context of JAX-WS web service development?

- A) SOAP Implementation Bridge
 - B) Service Integration Bean
 - C) Service Implementation Bean
 - D) SOAP Invocation Block
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7. A Service Implementation Bean (SIB) can be implemented as which two Java component types?

- A) Abstract class or Singleton class
 - B) POJO (Plain Old Java Object) or Stateless Session EJB
 - C) Servlet or JSP
 - D) Interface or Enum
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8. Which of the following is the correct `Endpoint.publish()` syntax for publishing a web service at port 9873?

- A) `Endpoint.deploy("http://localhost:9873/ts", TimeServerImpl.class);`
 - B) `Endpoint.start("http://localhost:9873/ts", "TimeServer");`
 - C) `Endpoint.publish("http://127.0.0.1:9873/ts", new TimeServer2Implementation());`
 - D) `Endpoint.register("9873/ts", new TimeServer2Implementation());`
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Section C: JAX-WS Annotations

9. Which annotation is used to mark a Java interface or class as a web service in JAX-WS?

- A) `@SOAPService`
 - B) `@WebService`
 - C) `@ServiceEndpoint`
 - D) `@JAXWSService`
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10. Which annotation exposes a specific method as an invokable web service operation?

- A) `@Expose`
 - B) `@RemoteMethod`
 - C) `@WebMethod`
 - D) `@ServiceMethod`
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11. What is the purpose of the `@SOAPBinding(style = SOAPBinding.Style.RPC)` annotation?

- A) Specifies that the service uses REST-style HTTP methods
 - B) Marks the class as a stateless session bean
 - C) Specifies that the SOAP messages use RPC (Remote Procedure Call) style binding
 - D) Registers the service with the UDDI registry
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12. In the `TimeServer2Implementation.java` code, what does `@WebService(endpointInterface = "JWS.TimeServer2Interface")` accomplish?

- A) Publishes the service at the JWS namespace URL
 - B) Links the implementation class to its SEI, forming the complete web service contract
 - C) Automatically generates the WSDL file for the service
 - D) Imports all methods from the `TimeServer2Interface` class
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Section D: Client Programming

13. In a JAX-WS Java client, what is the purpose of the `QName` object?

- A) To specify the SOAP security credentials
 - B) To identify the service by its namespace URI and service name as declared in the WSDL
 - C) To define the XML encoding format for messages
 - D) To set the maximum timeout for the web service call
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14. What is the correct sequence for building a JAX-WS Java client without `wsimport`?

- A) Get Port → Create QName → Load WSDL URL → Call Method
 - B) Load WSDL URL → Create QName → Create Service → Get Port → Call Method
 - C) Create Service → Load WSDL → Get Port → Create QName → Call Method
 - D) Import WSDL → Create Port → Bind to Service → Call Method
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15. In the `TimeServer` client, the WSDL URL used is:

- A) `http://localhost:9873/ts`
 - B) `http://localhost:9873/ts/service`
 - C) `http://localhost:9873/ts?wsdl`
 - D) `http://localhost:9873/wsdl/ts`
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16. A Perl client uses which module to consume a SOAP web service?

- A) `use HTTP::SOAP`
 - B) `use SOAP::Lite`
 - C) `use Webservice::Client`
 - D) `use XML::SOAP`
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Section E: wsgen and wsimport

17. What is the primary purpose of the `wsimport` tool?

- A) To generate JAXB classes from an annotated Java implementation class
 - B) To publish a web service endpoint on a specified port
 - C) To generate client stubs (SEI, Service class) from an existing WSDL file
 - D) To validate a WSDL document against the W3C specification
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18. Which development approach does `wsgen` follow?

- A) Top-Down (Contract-First)
 - B) Middle-Out
 - C) Schema-First
 - D) Bottom-Up (Code-First)
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19. What does the `-keep` flag do in a `wsimport` command?

- A) Keeps the WSDL file locally after import
 - B) Retains the generated Java source files instead of only the compiled `.class` files
 - C) Prevents overwriting existing source files
 - D) Keeps the web service endpoint running after generation
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20. Which flag in `wsimport` specifies the target Java package for the generated classes?

- A) `-t`
 - B) `-pkg`
 - C) `-p`
 - D) `-namespace`
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21. A client developer receives only the WSDL URL of a deployed service. Which tool should they use to build a Java client?

- A) `wsgen`
 - B) `javac`
 - C) `wsimport`
 - D) `wsdl2java` (non-standard)
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22. After running `wsimport`, which three types of artifacts are generated?

- A) SOAP envelope, HTTP headers, XML schema
 - B) Service Endpoint Interface (SEI), Service class, JAXB data beans
 - C) WSDL file, deployment descriptor, WAR archive
 - D) Annotation processor, binding file, endpoint publisher
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Section F: Document Style

23. What is the default SOAP binding style in JAX-WS if `@SOAPBinding` is omitted entirely?

- A) RPC style
 - B) Wrapped style
 - C) Document style
 - D) Literal style
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24. Which statement correctly distinguishes Document style from RPC style in JAX-WS?

- A) RPC style is the industry standard; Document style is only for legacy systems
- B) Document style sends full XML documents and supports complex data types; RPC style mimics a function call with simple types
- C) Document style is faster than RPC style because it uses JSON instead of XML
- D) RPC style supports both simple and complex types; Document style supports only simple types

25. In the Order Processing Service example, the `OrderClient2` uses `wsimport`-generated stubs. What is the first step in the 3-step client pattern when using generated stubs?

- A) Get the port by calling `service.getXxxPort()`
- B) Create a `QName` object with the service namespace
- C) Instantiate the generated Service class (e.g., `new OrderServiceImplService()`)
- D) Load the WSDL URL using `new URL("...")`

Answer Key

Q	Answer	Q	Answer	Q	Answer
1	C	10	C	19	B
2	D	11	C	20	C
3	B	12	B	21	C
4	C	13	B	22	B
5	B	14	B	23	C
6	C	15	C	24	B
7	B	16	B	25	C
8	C	17	C		
9	B	18	D		